

TOLCHESTER BEACH, MD, USA

WMO: 997312

Lat: 39.2135N

Lon: 76.2446W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.1	20.6	N/A	N/A	N/A	N/A	N/A	N/A	33.2	27.2	30.6	27.5	15.0	310	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	89.3	N/A	87.2	N/A	85.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	5.8	190

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Exterior Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		Extreme Max WB	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
30.0	25.4	21.7	DB	12.8	93.2	4.4	2.7	9.6	95.1	7.0	96.7	4.5	98.2	1.3	100.2	100.2
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual												
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.1	34.6	36.2	43.9	55.9	65.5	74.2	78.7	76.6	70.3	59.6	48.3	40.6
	DBStd	16.88	8.57	8.21	8.22	7.63	6.65	5.25	4.10	4.28	5.73	7.12	7.20	7.60
	HDD50	1560	482	393	226	27	0	0	0	0	0	9	116	307
	HDD65	4243	944	807	653	289	77	5	0	0	15	197	501	755
	CDD50	4167	3	7	38	203	482	726	891	823	608	305	64	17
	CDD65	1378	0	0	0	16	94	281	426	358	174	29	0	0
	CDH74	9904	0	0	1	93	405	1856	3843	2737	891	78	0	0
	CDH80	2511	0	0	0	17	61	408	1169	682	164	10	0	0
Wind	WSAvg	8.1	9.5	9.3	9.9	9.2	7.5	7.3	6.3	5.9	6.3	8.2	9.2	9.2
Precipitation	PrecAvg	44.9	3.5	2.7	4.3	3.6	3.7	3.9	4.3	3.8	4.6	3.7	3.1	3.5
	PrecMax	64.7	6.1	7.0	8.6	5.5	7.2	12.1	8.0	7.7	16.2	8.4	5.8	8.2
	PrecMin	33.0	1.3	0.3	0.0	1.7	0.9	1.3	1.7	0.7	0.9	0.2	0.9	1.2
	PrecStd	7.3	1.2	1.5	1.9	1.2	1.8	2.5	1.8	1.7	3.1	2.5	1.6	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	64.8	71.0	82.4	85.7	89.9	93.2	91.2	87.7	80.7	69.7	65.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	56.6	57.6	65.3	76.4	81.4	86.8	90.1	88.4	84.6	75.9	65.8	61.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	51.5	51.4	60.4	71.8	78.4	84.4	87.8	86.0	81.6	72.8	62.3	56.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	46.1	47.4	56.4	67.4	75.7	82.2	86.0	83.9	79.1	70.2	59.3	51.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.8	12.8	14.6	13.7	12.3	12.2	12.2	12.7	12.1	12.0	10.7
		MCDBR	19.9	17.9	20.5	20.2	15.5	13.9	13.5	13.3	14.3	14.1	15.9	16.0
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.312	0.320	0.366	0.400	0.447	0.499	0.518	0.513	0.426	0.363	0.327	0.310
	taud		2.469	2.435	2.359	2.290	2.194	2.092	2.052	2.051	2.269	2.451	2.508	2.517
	Ebn at Noon		273	287	282	278	266	252	245	242	259	267	266	266
	Edh at Noon		25	30	36	41	46	51	52	51	39	29	24	23
All-Sky Solar Radiation	RadAvg		648	920	1218	1566	1754	1940	1922	1701	1389	1026	740	568
	RadStd		53	104	94	125	211	148	91	99	157	110	63	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (76 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+236	+137

Nomenclature: See separate page